Assignment 3

1) Consider the following relational schema

branch (branch-name, branch-city, assets)
customer(customer-name,customer-street,customer-city)
account(account-number,branch-name,balance)
loan (loan-number, branch-name, amount)
depositor (customer-name, account-number)
borrower (customer-name, loan-number)

Write **relational algebraic expressions** for the following

1. Find all loans of over $1200
2. Find the loan number for each loan of an amount greater than $1200
3. Find the names of all customers who have a loan, an account, or both, from the bank
4. Find the names of all customers who have a loan and an account at bank.
5. Find the names of all customers who have a loan at the Perryridge branch
6. Find the names of all customers who have a loan at the Perryridge branch but do not have an account at any branch of the bank.
7. Find the names of all customers who have a loan at the Perryridge branch.
8. Find the largest account balance
9. Find all customers who have an account at all branches located in Brooklyn city.
10. Delete all account records in the Perryridge branch.
11. Delete all loan records with amount in the range of 0to 50
12. Delete all accounts at branches located in Needham

13. Insert information in the database specifying that Smith has $1200 in account A-973 at the Perryridge branch.
14. Update interest payments by increasing all balances by 5percent
15. Update all accounts with balances over $10,000, 6 percent interest and pay all others 5 percent
16. Create the view (named all-customer) consisting of branches and their customers.

2) Consider the following relational database, where primary keys are underlined.

employee(person_name,street,city)

works(person_name,company_name,salary)company(company_name,city)

manages(person_name,manager_name)

Given an expression in the **relational algebra** to express each of the following queries.

i) Find the name of all employees who work for first bank corporation.
ii) Find the name and cities of residence of all employees who work for first bank corporation.
iii) Find the name, street address and city of residence of all employees who work for first bank corporation and earn more than $10000 per annum.
iv) Find the name of all employees in this database who lives in same city as the company for which they work.
v) Find the name of all employees who live in the same city and on the same street as do their managers.
vi) Find the name of all employees in this database who do not work for first bank corporation
vii) Display the name of employee whose name begin from S.
viii) Find the average salary of employee
ix) Find the average salary of employee company wise
x) Find the name of all employees who earn more than their managers
xi) Find the name of employees who earn more than opearner at "NBL company" in the database.
xii) Find the name of all employee who live in "Lalitpur" and salary is less than 50000
xiii) Assume that companies may located in several cities. Find all companies located in every city in which small bank corporation is located.
xiv) List the name and city of employee who working "Pokhara"and have salary greater than Rs.50000
xv) Delete all employee who come from "Chitwan"
xvi) Increase salary of all employee by 15%
xvii) Insert new records in employee relation

xviii) Create view for which employee earns 20000$ or more

xix) Modify the database so that Ramesh now lives in Kathmandu

xx) Give all salary of employee of “ABC company” 18.5% rise

3) Differentiate between relational algebra and relational calculus? Define TRC and DRC?

4) Consider the following relational schema

Sailors (sid,sname,rating,age)

Boats(bid,bname,color)

Reserves(sid,bid,day)

Write a relational algebra expression for the query "find the name of sailors who have reserved red or green boat".

5) Create an un-normalized table and normalized it up to 3NF.

6)

i) Find the closure set of attributes and functional dependency from following $A \rightarrow BC, B \rightarrow C, AB \rightarrow C, AC \rightarrow D$

ii) Compute the closure of functional dependencies $AB \rightarrow C, B \rightarrow C, C \rightarrow D, D \rightarrow A$ for the relation $R=(A,B,C,D)$.Also list candidate key, prime and non-prime attributes.

iii) Let $R=\{A,B,C,D,E,F\}$ $F=\{A \rightarrow BC, E \rightarrow CF, B \rightarrow E, CD \rightarrow EF\}$Now compute $(AB)^+$

Normalization

i) Consider the following table. Identify all the functional dependencies in the table and convert it into 3NF.

Employee

EmpNo	EName	City	Post	Salary
E001	Ram	Pokhara, Dharan	Manager	30000
E002	Sita	Kathmandu	Analyst	25000
E003	Bharat	Bharatpur, Pokhara	Programmer	22000
E004	Laxman	Butwal	Manager	30000
E005	Dashrath	Lalitpur, Kathmandu	Analyst	25000

ii) Explain second Normal form? Whether the following table is in 2NF?

Sector	PlotNo	City
A	1	KTM
A	2	KTM
A	3	KTM
B	1	Pokhara
B	2	Pokhara

iii) Assume the un-normalized relation as given below and find the final normalized logical ER diagram normalizing the un-normalized relation up to 3NF explaining what you going to check at each normal step.

Roll.No	Name	SubID	SubName	FeePaid
1	HariManDangol	DBMS	DatabaseManagementSystem	20000
2	MohanPrasadShah	DBMS	DatabaseManagementSystem	20000
3	Indira Rimal	DBMS	DatabaseManagementSystem	30000
1	HariManDangol	CPROG	C programming	20000
2	MohanPrasadShah	CPROG	C programming	15000
3	Indira Rimal	MATH	Mathematics	30000